MIL-C-1313F(MU)
10 November 1971
SUPERSEDING
MIL-C-1313E(MU)
18 July 1968

MILITARY SPECIFICATION
CARTRIDGE, CALIBER .30, BALL, M2

1. SCOPE

1.1 This specification covers Cartridge, Caliber .30, Ball, M2 for use in Caliber .30 weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-45607	- Inspection Equipment Aquisition Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspections Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1168	- Lot Numbering of Ammunition

DRAWINGS

U.S. Army Munitions Command

C6137544	- Cartridge, Caliber .30, Ball, M2
D7553510	- Packing, and Marking; Cartridge, Cal. .30; Linked; Box, Ammunition, M19A1; Box, Wirebound
D7553698	- Packing and Marking; Cartridges, Cal. .30; 8 Round Clips; Bandoleers; Box, Ammunition, M19A1; Box, Wirebound

FSC 1305

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2.1 DRAWINGS (Cont'd)

D7553807	- Packing and Marking; Cartridges, Cal. .30; Cartons; Box, Ammunition, M2A1, Wirebound
D7553811	- Packing and Marking; Cartridges, Cal. .30; 5 Round Clips; Bandoleers; Box, Ammunition, M19A1; Box' Wirebound
D7553887	- Packing and Marking; Cartridges, Cal. .30; 5 Round Clips Bandoleers; Box, Ammunition, M2A1; Box, Wirebound
C7643674	- Classification of Cartridge Case Defects

PUBLICATIONS

TECP 700-700 VOL. III	- Manual of Test Methods for Small Arms Ammunition
AMCR 715-505 Volume 3	- Ammunition Ballistic Acceptance Test Methods, Volume 3, Test Procedures for 7.62-MM Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained form the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with the requirements specified on Drawing C6137544, referenced specifications and the following:

3.2 Bullet extraction.-The force required to extract the bullet from the cartridge case shall be not less than 45 pounds.

3.3 Residual stress.-The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Waterproof.-The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 15 seconds.

3.5 Accuracy.-The average of the mean radii or all targets of the sample cartridges at 600 yards shall be not greater than 7.5 inches.

3.6 Action time.-The action time (overall primer ignition, propellant burning and bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

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3.7 Velocity.- The average velocity of the sample cartridges shall be 2740 feet per second (ft/sec) plus or minus 30ft/sec at 78 ft. from the muzzle. The standard deviation of the velocities shall not exceed 32ft/sec.

3.8 Chamber pressure.-The average chamber pressure of the sample cartridges shall not exceed 50,000 (psi).

3.9 Stripping.-The jacket of the bullet or any part thereof, shall not separate from the core when the cartridge is fired.

3.10 Function and casualty.-The cartridge shall function without casualty and neither the neck of the cartridge case nor the chamber of the weapon shall show any evidence of deformation (peening) after firing.

3.11 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects.-The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.11.2 Foreign matter.-The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 2500 cartridges. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same sources of supply.

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4.2.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specification at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval. Determination as to acceptability of any initial production sample shall be based upon results of initial tests only, and no second tests shall be permitted on that initial production sample.

4.3 Inspection provisions.-

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.-Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number in accordance with MIL-STD-1168. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2. Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The AQL for Major Class shall be 0.25 percent and the AQL for the Minor Class shall be 1.50 percent.

4.3.2.1 Classification of defects.-The classification of defects shall be as specified in Table I.

TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded, or stained if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L or M location	X			
	in I, S or J location		X		

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4.3.2.1 (Cont'd)

TABLE I

No.	Defect and Method Inspection	Critical	Major	Minor	Major or Minor
	Case				
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Case mouth not crimped in cannelure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet jacket		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal (bullet)				X
28	Upset (crooked) point			X	
29	Exposed steel (clad jacket)			X	
30	Blunt point			X	
31	Defective cannelure			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing (primer pocket joint)			X	
38	Defective crimp			X	

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4.3.2.1 (Cont'd)

TABLE I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 20 pounds dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum, min.		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min. 2/	X			

1/ Refer to MIL-STD-636 for visual standards for defects 1 through 38.
2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain 25 grains or more of propellant shall be classed as a major defect. Any cartridge containing less than 25 grains of propellant shall be classified as a critical defect.

4.3.3. Tests.-The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.-The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridge selected shall be thoroughly mixed before being devided into samples for the various tests.

TABLE II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.2
Residual stress (mercurous nitrate) 1/	50	3.3
Waterproof 2/	20	3.4
Accuracy 3/	90	3.5

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4.3.3.1 (Cont'd)

TABLE II

Test	Number of Cartridges	Requirement Paragraph
Action time 1/	50	3.6
Velocity 3/	20	3.7
Chamber pressure 3/	20	3.8
Stripping 4/	100	3.9
Function and casualty 5/	--	3.10
Gun, Machine, Cal. .30, M37 (Tank) or Alternate Gun, Machine Cal. .30, Browning, M1919A4	300	
Gun, Machine, Cal. .30 M1919A6	300	
Rifle, U.S. Cal. .30 M1	408	

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the initial test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any defective cartridges are found in the second sample, the lot shall be rejected.

2/ Failure of nine or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than four but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

3/ Failure of the cartridges to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

4/ Any evidence of stripping as indicated by any irregular perforation on the paper screens, shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Any evidence of stripping in the second sample shall be cause for rejection of the lot.

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4.3.3.1 (Cont'd)

5/ The lot shall be rejected when function and casualty defects plus firing defects observed in all other firing tests exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first tests exceeds the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all service weapons specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred in the first test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

TABLE III

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Sample)
1. Misfire 4/		
a. No vent hole or obstruction in the vent area	0	-
b. Other	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup		
(1) Machine guns	12	28
(2) Rifles	2	4
b. Escape of gas through primer cup other than 3a	1	2
c. Escape of gas around primer cup more than 50% of periphery	10	21
d. Blown primer	0	1
e. Primer falls out of pocket on retraction of bolt	0	1
f. Primer remains in pocket but is physically loose	4	7

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4.3.3.2 (Cont'd) TABLE III

Defects	Acceptance Number First Sample	Acceptance Number Cumulative (First & Second Sample)
4. Case Casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I and S)	24	53
(2) Body (J)	4	7
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, shoulder or body (J and S)	1	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
5. Failure to extract	0	1
6. Weapon stoppage 3/	0	1
7. Deformation (peening) 1/	0	-

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

3/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests, shall be included.

4/ Each cartridge that misfires shall be disassembled and examined for presence of a vent hole in primer pocket, or any obstruction in the vent hole area of the primer pocket that can be assignable as the cause for misfire. If vent hole is missing or obstructed, the lot shall be rejected with no second sample permitted.

4.3.4 Packaging, packing and marking inspection.-During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy or blank shall be classed as a critical defect. Occurrence of any incorrect type other than those listed shall be classed as a major defect. All nonconforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

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4.3.5 Inspection equipment.-Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.-The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute.

4.4.2 Residual stress (Mercurous nitrate).-Tests shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.3 Waterproof.-The cartridges (not exceeding five at one time) shall be placed horizontally under a 2 to 2½ inch head of freshly boiled water in a vacuum jar test unit. The desiccator or jar shall be evacuated to a pressure of 5 pounds per square inch (10 inches of mercury) below atmospheric pressure and held at that pressure for not less than 15 seconds. The number of bubbles liberated from each bullet shall be observed.

4.4.4 Accuracy.-The test shall be conducted in accordance with TECP 700-700, Vol. III, utilizing the longest range available up to 600 yards. Ranges shorter than 200 yards shall not be used. When accuracy testing is conducted on rages shorter than 600 yards, the average of the mean radii of the targets shall be multiplied by the applicable conversion factor given below:

Range (yards)	200	300	400	500
Factor	3.40	2.23	1.68	1.35

4.4.5 Action time.-The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.6 Velocity.-The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.7 Chamber pressure.-The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.8 Stripping test.-The test shall be conducted in accordance with AMCR 715-505, Volume 3. Machine Gun M37 or alternate Machine Gun M1919A4 and Machine Gun M1919A6 shall be used, firing 50 round bursts in each gun.

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4.4.9 Function and casualty.-In these firings the weapon shall be at room temperature at the beginning of the test and the machine guns shall be cooled between bursts. The number of cartridges to be fired shall be as specified in Tabel II. The test shall be conducted in accordance with TECP-700-700, Vol. III and as indicated below:

a. Gun, Machine, Caliber .30, M37 (Tank) or alternate Gun, Machine, Caliber .30, Browning, M1919A4 - Fire in bursts of 100 cartridges.

b. Gun, Machine, Caliber .30, Browning, M1919A6 - Fire in bursts of 100 cartridges.

c. Rifle, U. S. Caliber .30, M1 - Fire 8 cartridges (one clip) rapid fire, with intervals of not more than ½ minute between clips. The weapon shall be cooled after firing the first 48 cartridges (6 clips). Two rifles shall be used and 104 cartridges shall be fired in each rifle.

d. Rifle, U. S. Caliber .30M1 - One hundred (100 cartridges in each of two rifles shall be fired single shot. After the cartridge has been chambered and the rifle bolt locked in the firing position, the rifle muzzle shall be elevated to a least a 45° angle above horizontal. The weapon shall then be lowered slowly to a horizontal position taking care that the muzzle of the weapon is not lowered below the horizontal position so that the propellant remains rear-most at the instant of firing.

4.4.9.1 Deformation (peening).-The expended cartridge cases and the chambers of the weapons shall be examined after the completion of the firing specified in (a), (b) and (c) above, and after each shot fired in (d) above, for any evidence of peening deformation. Any evidence of peening deformation of the weapon chamber or any evidence of a projection or localized expansion in the neck of the cartridge case shall be considered unacceptable.

4.4.10 Defect penalty.-In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing.-Level A (Worldwide shipment). The cartridges shall be packed in accordance with Drawings D7553510, D7553698, D7553811, D7553887, or D7553807.

5.2 Marking and labeling.-Packing boxes shall be marked and labeled in accordance with the applicalbe drawings as cited in paragraph 5.1.

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6. NOTES

6.1 Ordering data.-Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing

6.1.3 Provisions for supply, maintenance and disposition of the mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection Systems Requirements.

6.2 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize those hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

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Preparing activity:

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